MILITARY SPECIFICAION

MAT, SILHOUETTE LAYOUT

This specification is approved for use by Natick Research and Development Laboratories, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of tool layout mat.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-B-636	-	Boxes, Shipping, Fiberboard

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7690

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DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

5-7-188 - Mat, Silhouette Layout

(Copies of documents required by manufacturers in connnection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed by technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.4).

3.2 Materials. Materials shall conform to the requirements specified herein and in the subsidiary documents applicable thereto (see 6.5).

3.2.1 Base cloth. The base cloth shall be a 2.8 ounce per square yard, plain weave cloth made of 150 denier polyester yarns.

3.2.1.1 Color. The color of the base cloth shall be Army Tan No. 445 and shall match the standard sample (see 6.3).

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3.3 Construction.

3.3.1 Printing. The dyed base cloth shall be printed on one side only in the silhouette pattern of Drawing 5-7-188. The pattern shall be centered on the cloth. The color of the pattern shall be black. The full scale printed pattern shall depict the machinist's steel rule as 6 inches ± 1/8 inch long and the punch as 10 ± 1/8 inches long.

3.3.2 Coating. The dyed and printed base cloth shall be coated equally on both sides with a suitably compounded, clear, dull, thermoset polyurethane coating compound. If plasticizers are required, only phosphate or phthalate ester plasticizers shall be used. The finished, coated cloth shall meet all of the physical and chemical requirements shown in Table I when tested as specified in Table III.

TABLE I. Coated cloth requirements

Characteristic	Requirement	
	Min.	Max.
Weight, oz/yd^2	4.0	4.5
Breaking strength, lbs		
Warp	140	---
Filling	140	---
Tearing strength, g		
Warp	600	---
Filling	600	---
Blocking, rating	---	No. 2
Stiffness, warp only, cm		
at 70° ± 2°F	---	11.0
at -40° ± 5°F	---	20.0
Resistance to solvents, percentage change		
Acetone	---	3.0
Toluol	---	1.0
Abrasion resistance	The coating shall not be abraded from the coated cloth exposing the printed base cloth in the center 1 inch of the abraded portion.	

3.3.3 Size of mat. The cloth mat shall be 46 ± 1 inches long by 37 ± 1 inches wide.

3.4 Workmanship. The workmanship shall conform to the quality established by this document.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. The contractor shall furnish a certificate of compliance certifying that the base cloth conforms to the requirements of 3.2.1 and that the coating plasticizers, if used, conform to 3.3.2.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.1. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 End item inspection.

4.4.1.1 End item examination. The end item shall be examined in accordance with 4.4.1.1.1 and 4.4.1.1.2. The lot size shall be expressed in units of one mat. The sample unit shall be one mat.

4.4.1.1.1 Visual examination. The mats shall be examined for the defects listed in table II. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 major and 6.5 total (major and minor combined) defects per hundred units.

TABLE II. Visual defects

Examine	Defect	Classification Major	Minor
Cloth	Not color specified		X
	Torn	X	
	Wrinkled or lapped		X
	Soiled		X
	Edges frayed		X

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TABLE II. Visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Printing	Not color specified		X
	Not uniform density		X
	Incomplete		X
	Non-adherent	X	
Coating	Not coated both sides	X	
	Bare areas		X
	Uneven thickness		X
	Rough coated		X

4.4.1.1.2 Dimensional examination. The mats shall be examined for compliance with the dimensions specified. Any dimension not within specified requirements shall be classified a defect. The inspection level shall be II and the AQL shall be 6.5 defects per hundred units.

4.4.1.2 End item testing. The coated cloth shall be tested as specified in table III for conformance with the requirements of 3.3.2. The lot size shall be expressed in units of mats. The sample unit shall be one mat. The inspection level shall be II and the AQL shall be 6.5 defects per hundred units.

TABLE III. Test methods

Characteristics	Requirement paragraph	Test methods
Weight	3.2	5041 1/
Breaking strength	3.2	5100 1/
Tear strength	3.2	D 1424 2/
Blocking	3.2	5872 1/
Stiffness	3.2	5204 1/ 3/
Resistance to solvents	3.2	4/
Abrasion resistance	3.2	5304 1/ 5/

1/ Refers to FED-STD-191.

2/ Refers to ASTM.

3/ For low temperature tests, the specimens shall be subjected to a temperature of -40° ± 5°F for a minimum of 4 hours and the test shall be performed in a still atmosphere at that temperature.

4/ Five specimens of the coated cloth shall be weighed and then immersed completely in the chemical solvent for ten minutes. At the end of the exposure period, the specimens shall be removed from the solvent, air dried for 16 hours and then re-weighed. The percent weight change shall be calculated on the basis of the initial, unexposed specimen weight to the nearest 0.1 percent.

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5/ Three specimens of the coated cloth with long dimension parallel to the warp shall be abraded on the face side for 2000 cycles using Emery 0 polishing paper. The load shall be 3 pounds and the tension shall be 6 pounds.

4.4.2 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with section 5 requirements. Defects shall be scored as specified in table IV. The sample unit shall be one shipping container fully packaged and need not be closed. Examination for closure defects listed below shall be on shipping containers fully packaged. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE IV. Packaging defects

Examine	Defect
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, inadequate stapling, loose strapping, or improper taping. Bulged or distorted container.
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Content	Number of mats per container not as specified.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial (see 6.2).

5.1.1 Level A. Each mat shall be map-folded in quarters lengthwise and eights widthwise. The completely folded mat shall measure approximately 12 inches in length by 4-1/2 inches in width. Each mat shall be inserted into a snug-fitting polyethylene folder. The polyethylene folder shall be flat style, made from clear polyethylene film of 0.006 inch thickness (+ 20 percent tolerance). The folder shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the folder. One end of the polyethylene folder shall be open. The opening shall have a half moon shaped thumb notch cut out at the approximate center of the edge to allow for easy removal of the mat. The folder may be fabricated from polyethylene film, tubing or sheeting.

5.1.2 Commercial. Mats shall be preserved in accordance with MIL-STD-1188.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Fifty mats, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed on edge, 25 in length, one in width, and two in depth with a fiberboard pad between each layer within a shipping container. Inside dimensions of each shipping container shall approximate 19 inches in length, 12-1/4 inches in width and 9-1/2 inches in depth. Each shipping container shall be closed in accordance with method III and reinforced as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.2.

5.2.2 Level B packing. Fifty mats, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level B unit packs shall be packed on edge, 25 in length, one in width and two in depth with a fiberboard pad between each layer within a shipping container. Inside dimensions of each shipping container shall approximate 19 inches in length, 12-/14 inches in width and 9-1/2 inches in depth. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.2.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.2.

5.2.3 Commercial. Mats, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

5.3.1 Polyethylene folder marking. Each polyethylene folder shall be marked on one face with the item nomenclature in 1/2 inch high letters.

6. NOTES

6.1 Intended use. The Silhouette Layout Mat is to be used as a base on which the tools of a mechanic's automotive kit are displayed for field inspection.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) When a first article is required (see 3.1, 4.3, and 6.4).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Samples. Samples of the color shades may be obtained from the contracting activity or as directed by the contracting officer.

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long it meets the requirements of the document (see 3.2).

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MIL-M-44084(GL) Mat, Silhouette Layout

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